Logic Lecture 3 (16.10.2025)

Propositional logic – SYNTACTIC level.

1. SYNTAX — FORM

- vocabulary
- propositional variables (p, q, r ...)
- connectives

(!) $\neg$ negation → unary connective

(&&) $\wedge$ conjunction
(||) $\vee$ disjunction
(=>) $\rightarrow$ implication
(<=>) $\leftrightarrow$ equivalence

} → binary connectives

$\boxed{\text{order of precedence: } \neg > \wedge > \vee > \rightarrow > \leftrightarrow}$

- F_P : the set of formulas

2. SEMANTICS — MEANING

- semantic domain – the set of truth values

$\neg F = T$
$\neg T = F$

- new connectives – logic circuits later on.

$\uparrow$ „nand" – negation of „and" ($\wedge$)
$\downarrow$ „nor" – negation of „or" ($\vee$)
$\oplus$ „xor" – negation of „equivalence" ($\leftrightarrow$)

$$p \uparrow q := \neg(p \wedge q)$$
$$p \downarrow q := \neg(p \vee q)$$
$$p \oplus q := \neg(p \leftrightarrow q)$$

- Truth tables:

p, q → 2 variables → 2^2 lines in the table. (combinations)

		!	&&	||	=>	<=>	¬&&	¬||	¬<=>
p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$	$p \uparrow q$	$p \downarrow q$	$p \oplus q$

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \to q$	$p \leftrightarrow q$	$p \uparrow q$	$p \downarrow q$	$p \oplus q$
T	T	F	T	T	T	T	F	F	F
T	F	F	F	T	F	F	T	F	T
F	T	T	F	T	T	F	T	F	T
F	F	T	F	F	T	T	T	T	F

- Conjunction ($\wedge$) AND is T when ALL OPERANDS are T.
- Disjunction ($\vee$) OR is F when ALL OPERANDS are F.
- Implication ($\to$) is F when $\begin{cases} p \text{ is T} \\ q \text{ is F} \end{cases}$ $p \to q$.

  T cannot imply F.

- Equivalence ($\leftrightarrow$) is T when ALL OPERANDS HAVE THE SAME TRUTH VALUE.
- $\neq$ Exclusive or ($\oplus$) is T when ONE is T, ONE is F.

3. SUFFICIENCY VS. NECESSITY

p: $n \vdots 4$
q: $n \% 2 == 0$
r: $n \vdots 2$.

- x : If $\underbrace{n \vdots 4}_{p}$ then $\underbrace{n \% 2 == 0}_{q}$ $p \to q$

the premise is SUFFICIENT for the conclusion.

the conclusion is NECESSARY for the premise

- y : $\underbrace{n \% 2 == 0}_{q}$ if and only if $\underbrace{n \vdots 2}_{r}$ $q \leftrightarrow r$

the premise is NECESSARY and SUFFICIENT for the conclusion

4. CONDITIONAL RULES

IF – SUFFICIENCY

? ONLY IF – NECESSITY

$\boxed{\neg p \to \neg q \equiv q \to p}$ logically equivalent

$q \to p \equiv \neg p \to \neg q$

5. INTERPRETATION of a propositional formula

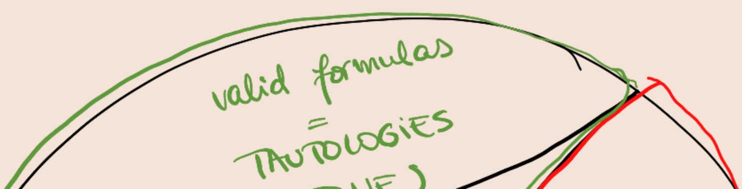
- interpretation: a function $i : \{p_1, \ldots, p_n\} \to \{F, T\}$
  $i : F_p \to \{F, T\}$

- $i(\neg p) = \neg i(p)$
- $i(p \to q) = i(p) \to i(q)$
- $i(p \wedge q) = i(p) \wedge i(q)$
- $i(p \vee q) = i(p) \vee i(q)$
- $i(p \leftrightarrow q) = i(p) \leftrightarrow i(q)$

- A formula → n variables → has 2^n interpretations.

6. SEMANTIC CONCEPTS

- model
  (i evaluates U as true => i is „a model of U".)
- anti-model
  (i evaluates U as false => i is „an anti-model of U").
- consistent formula
  (U has a model)
- valid formula = TAUTOLOGY → always true: $\models U$
  (U is true in all interpretations;
  all interpretations of U are models of U).
- inconsistent formula → always false → only F values in the table.
  (U does not have a model)
- contingent formula
  (consistent, but not valid)



$\models U$ metasymbol for tautology

without sth. on the left

$\models p \vee \neg p$ tautology

$\models \neg(p \wedge \neg p)$ contradiction

when it contains a pair of opposite literals

... $\equiv$ ... metasymbol for both true / both false

they have the same truth values. tables

$U \models V$ „V is a logical consequence of U".

↳ find the models of U;

these models have to be equal to the ones of V.

7. PROBLEMS IN PROPOSITIONAL LOGIC

1) Build the truth tables of the formulas

$U(p, q, r) \rightarrow 3$ propositions $\rightarrow 2^3$ lines in the table

$U(p,q,r) = (\neg p \vee q) \wedge (r \vee p)$

p	q	r	$\neg p \vee q$	$r \vee p$	$U(p,q,r)$

i1	T	T	T	T	T	T
i2	T	T	F	T	T	T
i3	T	F	T	F	T	F
i4	T	F	F	F	T	F
i5	F	T	T	T	T	T
i6	F	T	F	T	F	F
i7	F	F	T	T	T	T
i8	F	F	F	T	F	F

U has 4 models (i1, i2, i5, i7)

U has 4 antimodels (i3, i4, i6, i8)

$\underbrace{\qquad}_{\sim \text{ backtracking}}$

if all were T
$\Downarrow$
tautology

if all were F
$\Downarrow$
inconsistent

...

8. LOGIC EQUIVALENCES

1) Simplification laws
2) Commutative laws : $U \wedge V \equiv V \wedge U$
   $U \vee V \equiv V \vee U$
3) Distributive laws: $U \wedge (V \vee Z) \equiv (U \wedge V) \vee (U \wedge Z)$
   $U \vee (V \wedge Z) \equiv (U \vee V) \wedge (U \vee Z)$
4) Idempotency laws:
5) Absorbtion laws
6) De Morgan laws: $\neg(U \wedge V) \equiv \neg U \vee \neg V$
   $\neg(U \vee V) \equiv \neg U \wedge \neg V$
7) Associative laws: $(U \wedge V) \wedge Z \equiv U \wedge (V \wedge Z)$
   $(U \vee V) \vee Z \equiv U \vee (V \vee Z)$

! The T in a conjunction does not matter.

! The F in a disjunction does not matter.

...

dual logical equivalences

definition of connectives (a certain op. can be written as another op.)

Implication can be written as a neg + disj.

Any connection can be written as only $\uparrow$ or $\downarrow$.

9. SETS OF PROP. FORMULAS

1) consistent set $\{U_1, \dots, U_m\}$

$$i(U_1 \wedge U_2 \wedge \dots \wedge U_m) = T.$$

2) inconsistent set $\{U_1, \dots, U_m\}$

$$i(U_1 \wedge \dots \quad \wedge U_m) = F.$$

3) logical consequence of the set $\{U_1, \dots, U_m\}$

$U_1, U_2, \dots, U_m \models V$ → conclusion

$$i(U_1 \wedge U_2 \wedge \dots \wedge U_m) = T, \quad i(V) = T.$$

...

! A model of a hypothesis is also a model of a conclusion.

10. THEOREMS

T1

- $\models U \iff \neg U$ inconsistent
- $U \models V \iff \models U \rightarrow V \iff \{U, \neg V\}$ inconsistent
- $U \equiv V \iff \models U \leftrightarrow V$
- ...

T2 obvs

→ the example with the company (A, B, ...)

11. NORMAL FORM → prove consistency, validity...

- literal : $p, \neg p, q \dots$
- clause: → disjunction : $p, \neg p \vee q, \dots$
- ... → conjunction ...

- cube : → conjunction : $q,\ p \wedge \neg q$

- DNF = disjunction of cubes ⇒ disjunction of conjunctions

  ↳ all the models of the formula : $\underbrace{p}\vee\underbrace{\neg q}\vee\underbrace{r}$ 3 cubes

- CNF = conjunction of clauses ⇒ conjunction of disjunctions

  clauses ↓ disj.

  ↳ all the anti-models of the formula

$p \vee \neg p$ is T. → why

Property

12. NORMALISATION ALGORITHM

1) replace arrows with other operations
2) De Morgan
3) distribution

T3

T4

① DNF of the formula and find its models.

$$X = (p \wedge q \xrightarrow{1} r) \xrightarrow{2} (p \xrightarrow{3} r) \wedge q$$ → formula

! implication is the last connective

! $\boxed{p \to q \equiv \neg p \vee q}$

1) replace the arrows :

$$X \overset{1,3}{\equiv} (\neg(p \wedge q) \vee r) \xrightarrow{2} ((\neg p \vee r) \wedge q)$$

$$\overset{2}{\equiv} \neg(\neg(p \wedge q) \vee r) \vee ((\neg p \vee r) \wedge q)$$

2) De Morgan :

$$\equiv \underbrace{(p \wedge q) \wedge \neg r}_{\text{cube } \checkmark} \vee ((\neg p \vee r) \wedge q)$$

3) distribution :

$$\equiv \underbrace{(p \wedge r) \wedge \neg r}_{\text{cube 1}} \vee \underbrace{(\neg p \wedge q)}_{\text{cube 2}} \vee \underbrace{(r \wedge q)}_{\text{cube 3}}$$

=> disjunction of conjunctions => DNF w 3 cubes.

In order to find the model we need to ask ourselves:

" When is a disjunction True?

↳ cube 1 is true => model is true

↳ provides one true model

$i_1 : \{p, q, r\} \to \{T, F\}$ → if r was missing, we assign T/F to it.

$i_1(p) = T$

$i_1(q) = T$

$i_1(r) = F$

$\{p\} \to$ 4 models

! distinct models ($i2 = i5$)

anti-models?

Cube 1 → one model: $i_1(p) = T$

$i_1(q) = T$

$i_1(r) = F$

Cube 2 → 2 models: $i2(p) = F$ $i3(p) = F$

$i_2(q) = T$ $i_3(q) = T$

$i_2(r) = T$ $i_3(r) = F$

Cube 3 $\rightarrow$ 2 models: $i_4(p) = T$ $i_5(p) = F$

$i_4(q) = T$ $i_5(q) = T$

$i_4(r) = T$ $i_5(r) = T$

$i_2 = i_5$

The models of X: i_1, i_2, i_3, i_4.

The rest are anti-models of X.